**Project Design Phase-II**
**Technology Stack (Architecture & Stack)**

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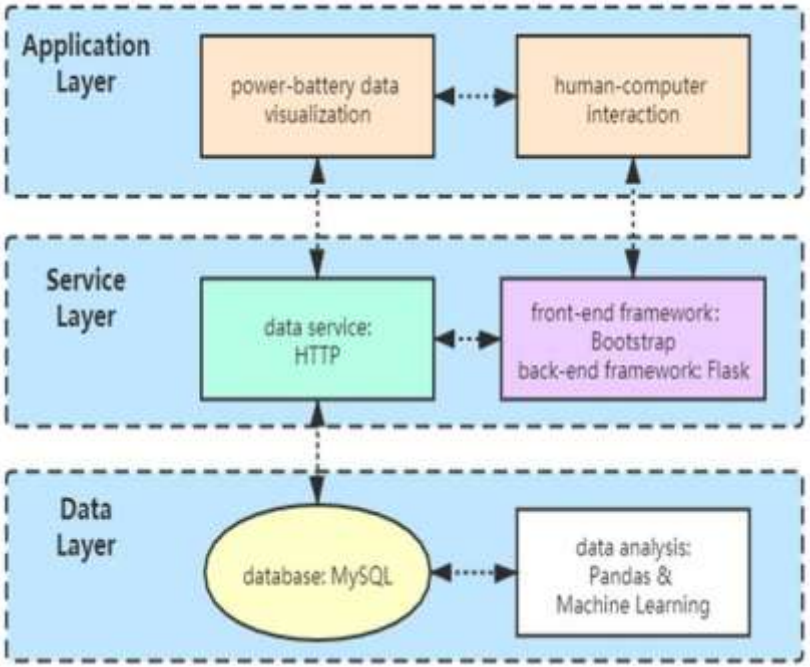
| Date | 25 February 2026 |
| Team ID | LTVIP2026TMIDS44140 |
| Project Name | Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau |
| Maximum Marks | 4 Marks |

**Technical Architecture:**




Guidelines:

The system processes housing data using Python preprocessing, Tableau visualization, and Flask web application.

Dataset and preprocessing run on local system, while dashboards are hosted on Tableau Public Cloud.

External interface used is Tableau Public and its embed visualization service.

Data is stored in local CSV dataset and Tableau dashboard cloud storage.

**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | User interacts with dashboard via website | HTML, CSS, Bootstrap, JavaScript |
| 2. | Application Logic-1 | Website backend logic | Python<br>Flask |

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|---|---|---|---|
| 3. | Application Logic-2 | Data preprocessing logic | Python (Pandas, NumPy) |
| 4. | Application Logic-3 | Data Visualization Engine | Tableu |
| 5. | Database | Housing dataset storage | CSV Dataset (Local Storage) |
| 6. | Cloud Database | Not Used (Optional Future Scope) | Tableau Public Cloud Storage |
| 7. | File Storage | Dataset storage | Local File System |
| 8. | External API-1 | Tableau Dashboard Embed Service | Tableau Public API |
| 9. | External API-2 | Not Used | — |
| 10. | Machine Learning Model | Not Used | — |
| 11. | Infrastructure (Server / Cloud) | Application Hosting | Local Server / Flask Server |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | Used open-source web and analytics tools | Flask, Bootstrap, Pandas |

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|---|---|---|---|
| 2. | Security Implementations | Basic access control and secure hosting | HTTPS, Flask Security Config |
| 3. | Scalable Architecture | Can scale by moving dataset to cloud | Cloud Hosting, Tableau Cloud |
| 4. | Availability | Dashboard available via web browser | Flask Web Server |
| 5. | Performance | Fast dashboard loading via Tableau Public | Tableau Optimization, Local Processing |